Two New Transiting Planet Candidates from an Independent TESS Search

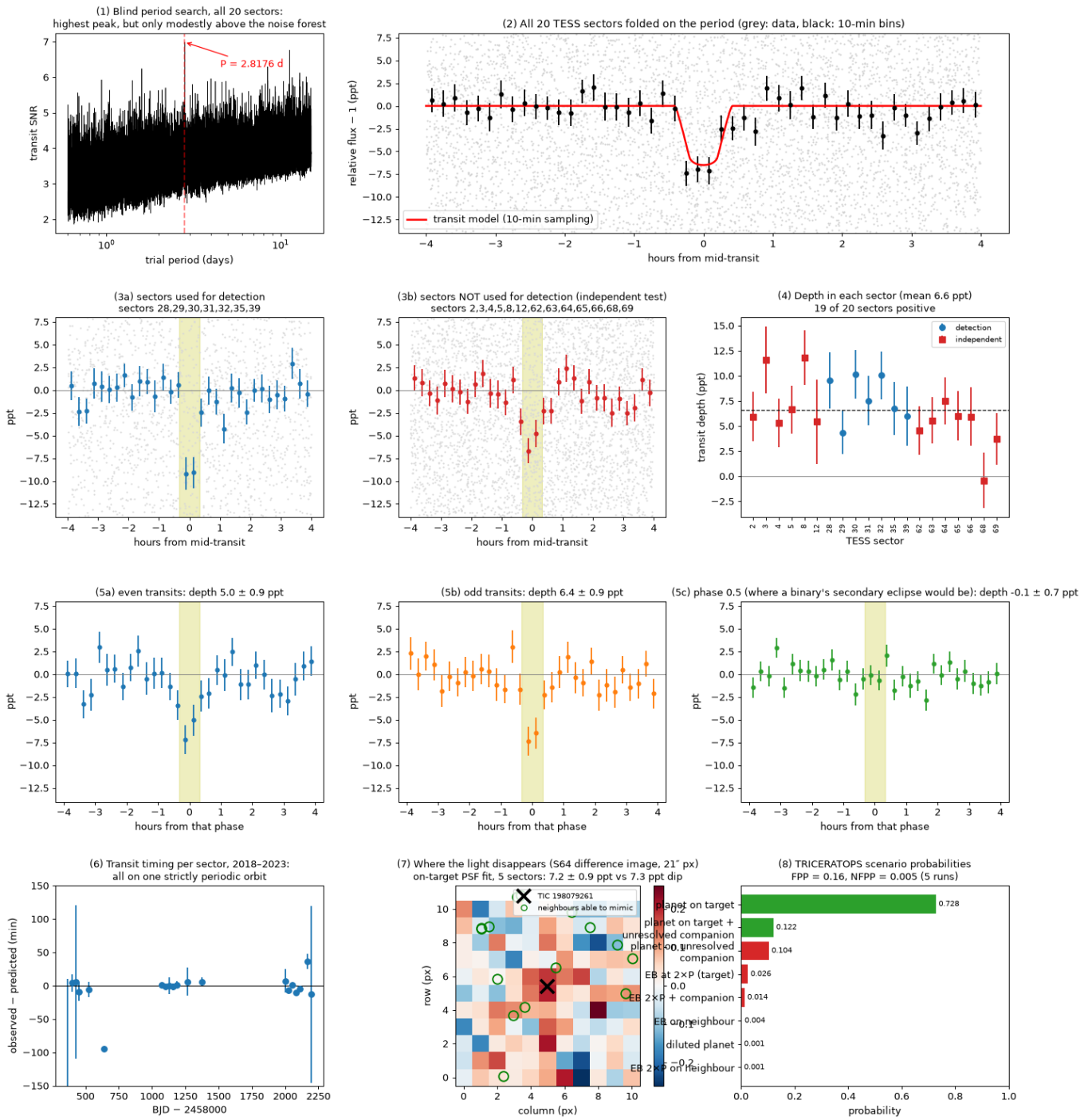
Term project, General Astronomy, Korea Science Academy of KAIST · Sarang Park (박사랑) · October 2026

Summary

	TIC 198079261 (primary)	TIC 366760231
RA, Dec (J2000)	04h16m06.4s −59°35′39″	22h20m28.3s +74°27′47″
Host star	Ultracool M6–M7 dwarf, 0.123 R☉, 31.6 pc, T = 15.0	M dwarf, 0.389 R☉, 55.3 pc, T = 12.4
Period	2.817636 ± 0.000018 d	3.137321 ± 0.000003 d
Mid-transit (BJD_TDB)	2460071.6821 ± 0.0024	2458962.04155 ± 0.00061
Depth / duration	≈0.65–0.73 % / 0.66 h	≈0.16–0.19 % / ≈0.55 h
Planet radius	1.02 ± 0.13 R⊕	≈1.9–2.5 R⊕
Equilibrium temperature	≈355 K	≈600 K
Independent-sector recovery	13 unused sectors, ≈4.9σ	3 unused FFI sectors, ≈6σ
TRICERATOPS FPP / NFPP	0.16 / 0.005	0.26 / <0.001
Known in TOI / CTOI / TCE / literature?	No (checked Oct 2026)	No (checked Oct 2026)

Both objects are **planet candidates**, not confirmed planets. Confirmation requires ground-based transit photometry, high-resolution imaging and spectroscopy. The search pipeline, vetting and figures were developed with the assistance of an AI tool (Claude).

TIC 198079261: evidence sheet



(1) blind period search; (2) all 20 sectors folded with the transit model; (3a/3b) detection vs independent sectors; (4) depth per sector; (5) odd/even transits and phase 0.5; (6) transit timing; (7) difference image (where the light disappears); (8) TRICERATOPS scenario probabilities.

TIC 198079261: a candidate Earth-sized planet on a 2.82-day orbit around a nearby ultracool (M6–M7) dwarf

Status: new planet candidate from an independent search of public TESS full-frame-image (FFI) light curves. Not confirmed. Checked on 2026-10-21. The signal is not a TOI, a CTOI or a confirmed planet. It is not in any SPOC 2-min TCE catalog (132 runs through sector 96), and it is not a TESS-SPOC FFI TCE: the star has no data-validation products in any of the 51 TESS-SPOC DV releases. It is not in the public candidate lists of *ahrazzle/exoplanets* or *tess-transit-hunter*. A web search on its TIC, 2MASS and Gaia identifiers returns nothing. ExoFOP lists no follow-up observations.

Host star (TIC v8.2 / Gaia DR3)

IDs	TIC 198079261, 2MASS J04160640−5935386, Gaia DR3 4678628654309752576, WISE J041606.56−593537.5
RA, Dec (J2000)	64.02671, −59.59404; pm 116.7, 93.5 mas/yr
TESS / Gaia G / K mag	15.03 / 16.66 / 12.02
Teff, R★, M★	2825 ± 157 K, 0.123 ± 0.004 R☉, 0.099 ± 0.020 M☉ (ultracool dwarf, ~M6–M7)
Distance	31.55 ± 0.09 pc (Gaia parallax 31.64 mas)
Gaia RUWE / non_single_star	1.02 / 0, so no astrometric sign of binarity
TIC contamination ratio	0.007

Candidate parameters

MCMC (batman + emcee) on unbinned TESS-SPOC 200-s data from S62–S69, with a stellar-density prior from the TIC. The ephemeris comes from a linear fit to per-sector mid-times over 20 sectors (2018–2023), with uncertainties scaled by √(χ²/dof).

Parameter	Value
Period	**2.817636 ± 0.000018 d**
Mid-transit epoch	**2460071.6821 ± 0.0024 BJD_TDB**
Depth (model, at mid-transit)	6520 ± 1440 ppm (box depth over all sectors: 7314 ppm)
Duration T14	0.66 ± 0.08 h
Rp/R★	0.076 ± 0.009
Impact parameter	0.44 (+0.23/−0.28)
a/R★	31.6 ± 2.3
Planet radius	**1.02 ± 0.13 R⊕**
Teq (A = 0), insolation	**355 ± 13 K, 2.7 ± 0.4 S⊕**
Detection SNR (all sectors)	13.3
Expected RV semi-amplitude	~2–3 m/s for ~1 M⊕ (0.1 M☉ host)

Stellar density from the transit shape alone: a fit without the density prior gives ρ★ = 90 (+64/−66) g cm^{-3}, consistent with the TIC value of 75 ± 17. In other words, the transit shape and duration fit this specific ultracool host.

Evidence

Test	Result	Figure
Discovery	GPU BLS with rotation prewhitening, flare masking and bad-time masking, on TESS-SPOC FFI sectors 28–39. The vetter v3 ensemble ranked it at the top of the unknown signals (score 1.00).	—
Independent confirmation	13 sectors not used for detection (S2–12 at 30-min cadence, S62–69 at 200-s), at the detection-only ephemeris with the ±3.3 h timing window scanned. Depth 5314 ± 824 ppm. 0 of 150 identical scans at random phases reach this depth (random mean 2902 ± 495 ppm, ~4.9σ).	fig2
Depth consistency	19 of 20 sectors show a dip of 4900–12700 ppm, each consistent with ~7–8 ppt. S68 shows 500 ± 3300 ppm (2.5σ low).	—

Test	Result	Figure
Odd vs even	6936 ± 775 vs 7696 ± 779 ppm (0.7σ)	—
Secondary eclipse	None detected	—
Pixel-level location	The stacked difference images over 5 TESS-SPOC FFI TPF sectors, fitted with the target's own PSF, give **7195 ± 903 ppm (8.0σ)** at the target position. That equals the light-curve depth, so the full dip comes from the target's PSF.	—
Gaia neighbours	The nearest star able to mimic the dip (G = 18.6, needing a 5% eclipse) is 26″ away (1.2 TESS px). Every capable neighbour is ≥ 26″ away.	gaia_neighbors_120arcsec.csv
TRICERATOPS (Giacalone et al. 2021)	**FPP = 0.159 ± 0.008, NFPP = 0.005** (5 runs). P(TP) ≈ 0.73. The remainder is mostly PTP/STP (an unresolved companion); EB scenarios total ≈ 0.04. The formal "likely planet" class needs NFPP < 0.001. The pixel-level on-target test above, which TRICERATOPS does not use, independently argues against nearby-star scenarios.	triceratops_probs.csv

Caveats

- The star is faint (T = 15.0). The detection relies on FFI photometry, and the per-sector SNR is only ~3–4.
- Per-sector mid-times scatter more than their formal errors (χ^2/dof = 5.1). Ephemeris uncertainties are inflated accordingly.
- An unresolved bound companion within ~1″ cannot be excluded without high-resolution imaging. This is the main source of the remaining FPP.
- Stellar parameters come from TIC v8.2 (photometric relations) and have not been spectroscopically confirmed.

Why it matters, if confirmed

An Earth-sized (1.0 R⊕), temperate-to-warm (~355 K) planet transiting a 0.12 R☉ ultracool dwarf at 31.6 pc would be a TRAPPIST-1-like system. The ~0.7% transit depth is deep enough for transmission spectroscopy with JWST.

Suggested follow-up

1. **Ground-based transit photometry in the red/near-IR** (e.g. SPECULOOS, TRAPPIST, MEarth-South, or a 1–2 m telescope in I/z) at a predicted time (upcoming_transits.md, ±14 min). A ~0.7% dip on an I ≈ 14–15 star is readily detectable from the ground. It would confirm the source at arcsecond resolution and refine the ephemeris.
2. High-resolution imaging (AO/speckle) to rule out companions within ~1″.
3. NIR spectroscopy to confirm the spectral type, then NIR RVs (K ≈ 2–3 m/s).

Reporting route

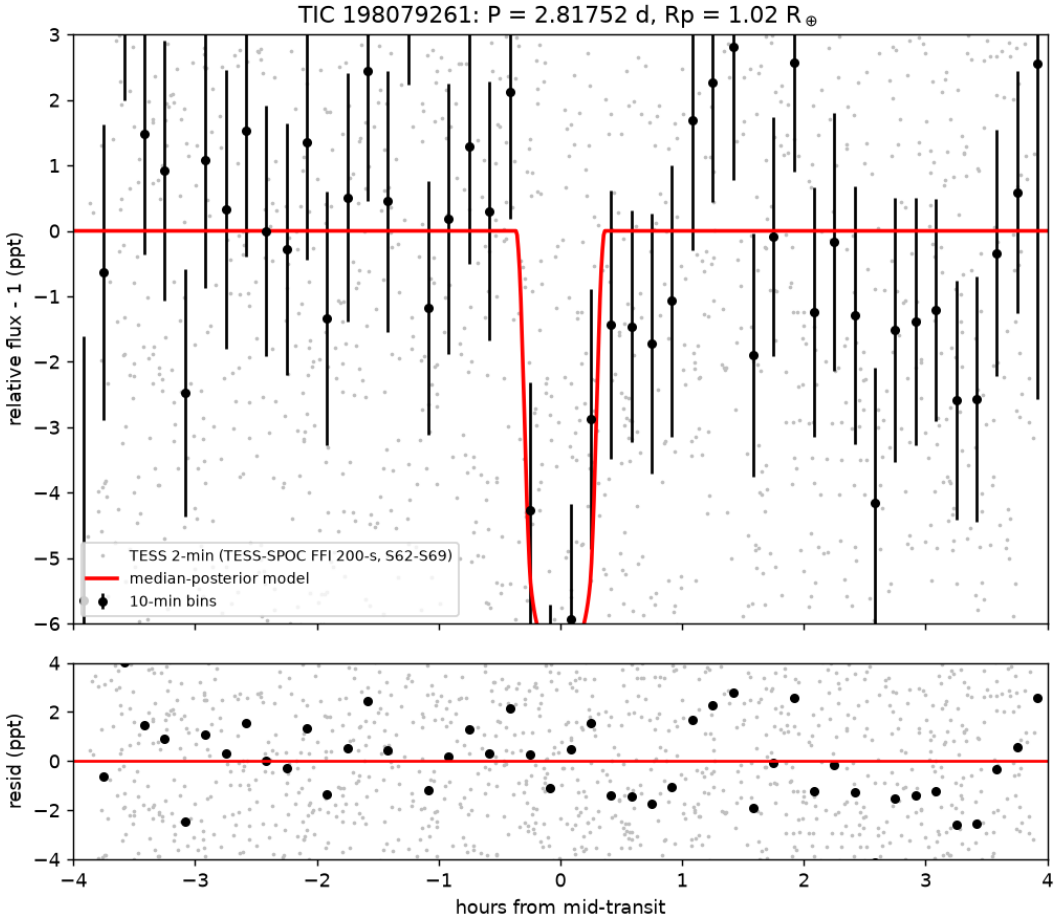
As with the other candidates: ExoFOP accepts only published community candidates (a refereed journal or an AAS Research Note citing published methods), or the target can be raised with the TESS TOI working group / exofop-support@ipac.caltech.edu.

Files

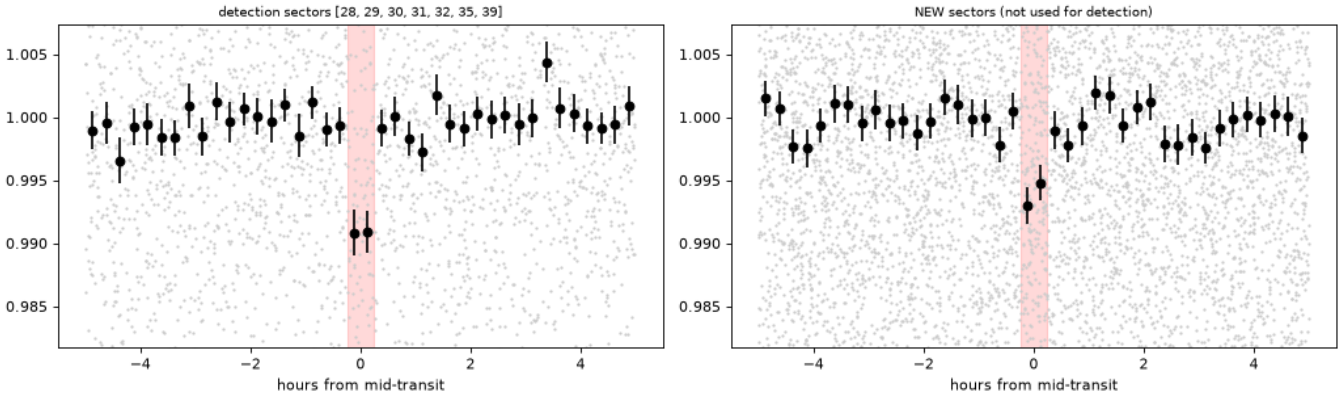
File	Contents
fit_results.json, fit_results_freerho.json, chain*.npy	MCMC fits, with and without the density prior
ephemeris.json, upcoming_transits.md	Linear ephemeris (scaled errors); predicted transits
fig1_transit_fit*.png	Folded 200-s data with model
fig2_detection_vs_independent_sectors.png	Detection sectors vs the 13 independent sectors

File	Contents
triceratops_probs.csv, vet_summary.json	Statistical validation and on-target pixel test
gaia_neighbors_120arcsec.csv	Gaia DR3 neighbours

Code: ~/tess_search/scripts/ (fetch_ffi.py, search_core.py, ffi_deep.py, vet_198079261.py, fit_198079261.py, ephem_198079261.py, train_vetter3.py).



Transit fit (TESS-SPOC FFI 200-s data, sectors 62–69).



Detection sectors vs the 13 independent sectors.

TIC 366760231: a new candidate ~2 R⊕ planet on a 3.14-day orbit around a nearby M dwarf

Status: new planet candidate from an independent search of public TESS data. Not confirmed. Checked on 2026-10-21. The signal is not a TOI, a CTOI or a confirmed planet. It is not in any of the 132 SPOC TCE catalogs (single- and multi-sector, through sector 96). It is not in the public candidate lists of the independent projects *ahrazzle/exoplanets* (~530 candidates) or *tess-transit-hunter*. A web search on its TIC, 2MASS and Gaia identifiers returns nothing. ExoFOP lists no follow-up observations for this star.

Host star (TIC v8.2 / Gaia DR3)

IDs	TIC 366760231, 2MASS J22202826+7427466, Gaia DR3 2279461636455560576, UCAC4 823-025985
RA, Dec (J2000)	335.11793, +74.46294; pm 129.6, 30.5 mas/yr
TESS / Gaia G mag	12.39 / 13.56
Teff, R★, M★	3456 ± 157 K, 0.389 ± 0.012 R☉, 0.379 ± 0.020 M☉ (early–mid M dwarf)
Distance	55.3 pc
Gaia RUWE	1.20
TIC contamination ratio	0.107

Candidate parameters

Ephemeris: linear fit to per-sector mid-transit times from all 9 sectors (2019–2025).

Parameter	Value
Period	**3.137321 ± 0.000003 d**
Mid-transit epoch	**2458962.04155 ± 0.00061 BJD_TDB**
Depth	1640 ± 220 ppm (free-density fit); 1915 ± 340 ppm (catalog-density fit, at mid-transit)
Duration T14	0.54–0.60 h (± 0.04–0.07)
Impact parameter	High: 0.97 (+0.23/−0.04) with the catalog stellar density
Planet radius	About 1.9–2.5 R⊕. The best catalog-density model has Rp/R★ = 0.045 (1.9 R⊕) at b = 0.93. The posterior tail to larger radii comes from the near-grazing degeneracy.
a/R★ (catalog density)	16.7 ± 0.6
Teq (A = 0), insolation	About 600 K, about 21 S⊕
Detection	SNR 9.4 (all sectors combined); transit $\Delta\chi^2 \approx 70$ relative to no transit
Expected RV semi-amplitude	About 3 m/s for ~4 M⊕

Note on the short duration: a central transit would last about 1.4 h. The observed ~0.55 h therefore needs a high impact parameter. A fit with free stellar density prefers a box-like shape with an implied $\rho_\star \approx 144$ g cm^{-3}. That fit is only marginally better ($\Delta\chi^2 = 2.5$), so the catalog-density, high-b solution is fully consistent with the data.

Evidence

Test	Result	Figure
Discovery	GPU box-least-squares search with rotation prewhitening, bad-time masking and flare-aware cleaning, on 2-min sectors 17, 18, 19, 24 and 25	—
Independent confirmation in data not used for detection	TESS-SPOC FFI sectors 58, 59 and 85 (2022–2024) at the ephemeris from the detection sectors (±1 h timing window scanned). Depth 1202 ± 233 ppm. 0 of 300 identical scans at random phases reach this depth (random mean 569 ± 101 ppm, ~6σ).	fig2
Sector 52 (2-min, also unused for detection)	1350 ± 455 ppm	fig3
Depth consistency	All 9 sectors positive and mutually consistent: 941–1883 ppm	fig3
Transit timing	9 per-sector mid-times over 6 years fit a linear ephemeris ($\chi^2 = 12.8/7$)	—

Test	Result	Figure
Period aliases	Best alias SNR 8.4 vs 5.0 for the runner-up (detection data)	—
Secondary eclipse	−93 ± 222 ppm (none)	—
Pixel-level location	Too shallow for single-sector difference-image centroids (S/N 1–2). Fitting the stacked difference images with the target's PSF gives 855 ± 178 ppm at the target position (4.8σ).	fig4
TRICERATOPS (Giacalone et al. 2021)	**FPP = 0.264 ± 0.017, NFPP = 0.0000** (5 runs). P(TP) ≈ 0.64; the remainder is mostly STP (planet on an unresolved companion) and PTP/EBx2P. All nearby-star scenarios are ≈ 0. This meets the "likely planet" criterion (FPP < 0.5, NFPP < 0.001) but not validation (FPP < 0.015).	triceratops_probs.csv

Caveats

- The SNR is modest (~9). Single-sector difference images cannot localize the signal.
- The duration is short for the host star and needs a high impact parameter. Near-grazing geometry leaves the radius poorly constrained and keeps unresolved-companion scenarios open.
- Nearest Gaia neighbour: G = 16.3 at 11.9″ (it would need a 2–3% eclipse). TRICERATOPS assigns it negligible probability, but direct confirmation needs higher angular resolution.
- QLP light curves for sectors 57, 77–79, 84 and 86 were missing or unreadable, so those sectors are untested.
- Stellar parameters come from TIC v8.2 and have not been spectroscopically confirmed.

Suggested follow-up

1. **High-resolution imaging** (speckle or AO). Ruling out companions within ~1″ is what would most reduce the FPP, toward validation.
2. **Ground-based transit photometry** at a predicted time (upcoming_transits.md, ±3 min). 1–2″ seeing resolves the 11.9″ neighbour and tests on-target depth and chromaticity. The depth (~1.6–1.9 ppt) is within reach of 0.6–1 m telescopes for a G = 13.6 star.
3. Reconnaissance spectroscopy to rule out a spectroscopic binary, then precise near-infrared RVs (K ≈ 3 m/s).

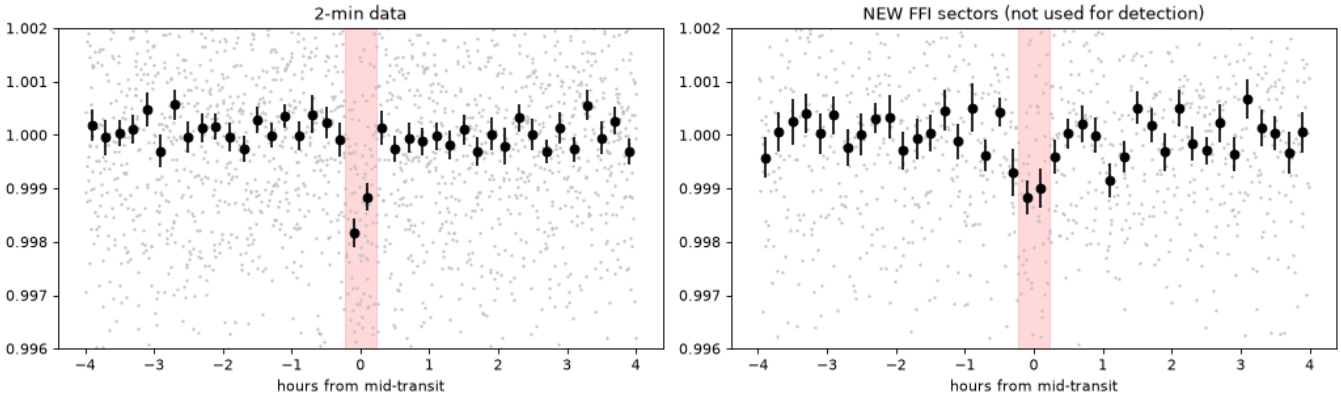
Reporting route

ExoFOP requires community candidates to be published, either in a refereed journal or in an AAS Research Note that cites previously published methods. The methods used here are all standard: BLS (Kovács et al. 2002); transit model (Mandel & Agol 2002; batman, Kreidberg 2015); emcee (Foreman-Mackey et al. 2013); TRICERATOPS (Giacalone et al. 2021); TESS-SPOC FFI light curves (Caldwell et al. 2020). Another option is to contact the TESS TOI working group or exofop-support@ipac.caltech.edu.

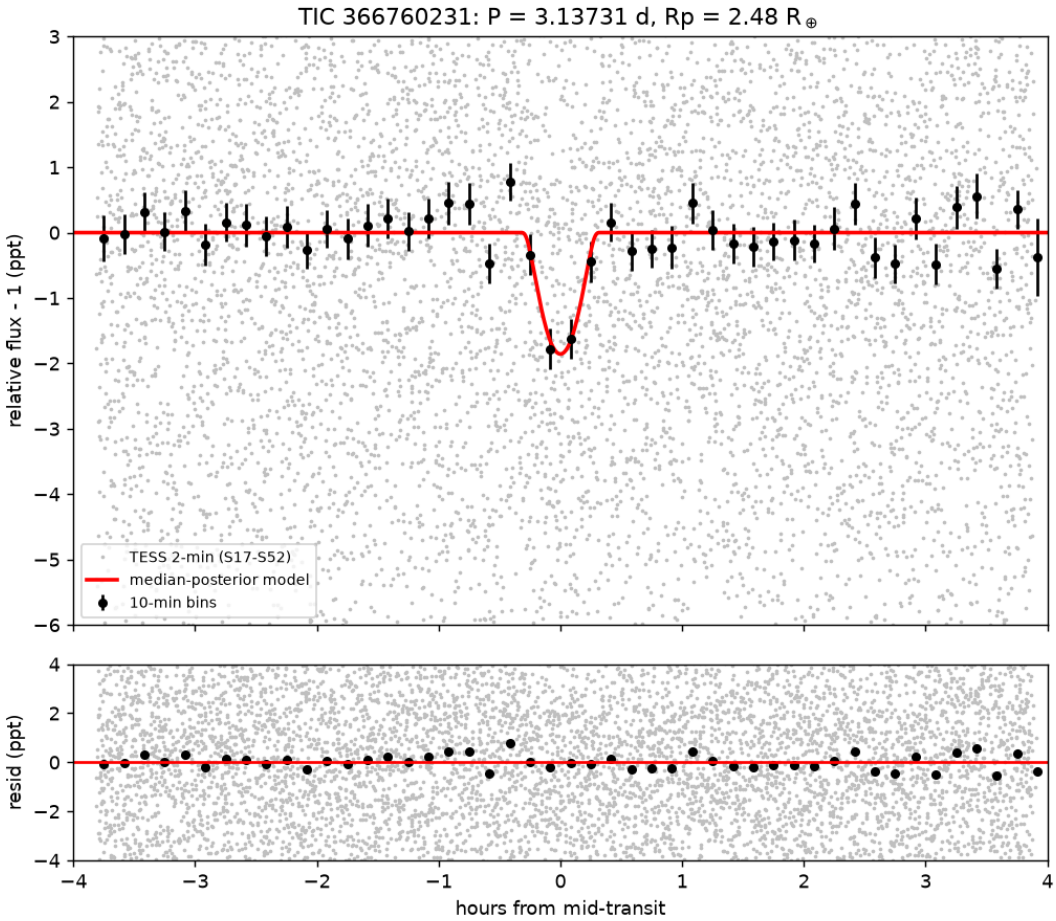
Files

File	Contents
fit_results.json, fit_results_freerho.json, chain*.npy	MCMC fits, with and without the stellar-density prior
ephemeris.json, upcoming_transits.md	Linear ephemeris with covariance; predicted transits
fig1_transit_fit*.png	Folded 2-min data with model
fig2_independent_FFI_sectors.png	Detection data vs the independent FFI sectors
fig3_per_sector_folds.png, fig4_difference_images.png	Per-sector folds; difference images
triceratops*.json, triceratops_probs.csv	Statistical validation runs
gaia_neighbors_120arcsec.csv	Gaia DR3 neighbours

Code: ~/tess_search/scripts/ (search_core.py, deep_vet.py, extra_sectors_366760231.py, fit_366760231.py, tric_366760231.py, ephem_366760231.py).



Detection data (2-min) vs independent TESS-SPOC FFI sectors 58, 59, 85.



Transit fit (2-min data, sectors 17–52).

Predicted transits

TIC 198079261 (observable from the southern hemisphere, e.g. KMTNet/SPECULOOS)

epoch	BJD_TDB mid-transit	date (approx.)	1σ
449	2461336.8007	2026-10-23 07:13 UTC	±14 min
450	2461339.6184	2026-10-26 02:50 UTC	±14 min
451	2461342.4360	2026-10-28 22:27 UTC	±14 min
452	2461345.2536	2026-10-31 18:05 UTC	±14 min
453	2461348.0713	2026-11-03 13:42 UTC	±14 min
454	2461350.8889	2026-11-06 09:20 UTC	±14 min
455	2461353.7065	2026-11-09 04:57 UTC	±14 min
456	2461356.5242	2026-11-12 00:34 UTC	±14 min
457	2461359.3418	2026-11-14 20:12 UTC	±14 min
458	2461362.1594	2026-11-17 15:49 UTC	±14 min
459	2461364.9771	2026-11-20 11:27 UTC	±14 min
460	2461367.7947	2026-11-23 07:04 UTC	±14 min
461	2461370.6124	2026-11-26 02:41 UTC	±14 min
462	2461373.4300	2026-11-28 22:19 UTC	±14 min

TIC 366760231 (circumpolar from Korea)

epoch	BJD_TDB mid-transit	date (approx.)	1σ
757	2461336.9933	2026-10-23 11:50 UTC	±3 min
758	2461340.1306	2026-10-26 15:08 UTC	±3 min
759	2461343.2680	2026-10-29 18:25 UTC	±3 min
760	2461346.4053	2026-11-01 21:43 UTC	±3 min
761	2461349.5426	2026-11-05 01:01 UTC	±3 min
762	2461352.6799	2026-11-08 04:19 UTC	±3 min
763	2461355.8173	2026-11-11 07:36 UTC	±3 min
764	2461358.9546	2026-11-14 10:54 UTC	±3 min
765	2461362.0919	2026-11-17 14:12 UTC	±3 min
766	2461365.2292	2026-11-20 17:30 UTC	±3 min
767	2461368.3665	2026-11-23 20:47 UTC	±3 min
768	2461371.5039	2026-11-27 00:05 UTC	±3 min
769	2461374.6412	2026-11-30 03:23 UTC	±3 min
770	2461377.7785	2026-12-03 06:41 UTC	±3 min